

LAFAYETTE, LA, USA

WMO: **722405**

Lat: **30.205N** Lon: **91.988W** Elev: **12** StdP: **101.19** Time Zone: **-6.00 (NAC)** Period: **94-19** WBAN: **13976**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	-1.1	0.9	-8.8	1.8	2.1	-6.0	2.3	4.9	10.4	14.1	9.4	13.7	3.2	10	0.386	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	8	9.1	35.0	25.4	34.0	25.3	33.1	25.2	27.1	31.5	26.7	31.1	26.3	30.7	3.1	50	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	26.1	21.5	28.6	25.6	20.9	28.4	25.2	20.3	28.2	85.0	31.5	83.1	31.1	81.5	30.7	31.8	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	9.0	8.1	7.2	DB	-4.2	36.6	2.3	1.3	-5.8	37.5	-7.2	38.3	-8.5	39.1	-10.2	40.0	(4)
(5)				WB	-5.5	28.3	2.1	1.3	-7.0	29.3	-8.3	30.1	-9.4	30.8	-11.0	31.8	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	20.7	11.5	13.7	17.0	20.6	24.7	27.4	28.3	28.4	26.2	21.4	16.1	12.7	(6)
(7)			DBStd	7.23	5.68	5.35	4.64	3.67	2.70	1.75	1.47	1.65	2.46	4.01	4.67	5.45	(7)
(8)			HDD10.0	120	51	23	6	0	0	0	0	0	0	0	7	34	(8)
(9)			HDD18.3	771	219	146	81	20	1	0	0	0	1	18	96	190	(9)
(10)			CDD10.0	4026	98	126	222	319	456	522	567	571	487	353	190	116	(10)
(11)			CDD18.3	1636	9	15	38	89	198	272	309	312	237	113	29	14	(11)
(12)			CDH23.3	15231	14	38	168	614	1738	2758	3305	3342	2166	883	160	43	(12)
(13)			CDH26.7	5982	0	1	11	112	612	1184	1418	1500	872	256	17	1	(13)
(14)	Wind		WSAvg	3.0	3.6	3.8	3.6	3.5	3.1	2.5	2.2	2.1	2.4	2.6	2.9	3.3	(14)
(15)	Precipitation		PrecAvg	1562	143	111	100	126	127	172	153	146	134	119	108	122	(15)
(16)			PrecMax	2167	339	232	270	303	342	455	236	708	262	352	350	342	(16)
(17)			PrecMin	1027	20	18	8	12	7	22	45	37	50	5	30	58	(17)
(18)			PrecStd	288	92	60	60	74	86	94	47	122	60	99	75	67	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		24.9	26.1	27.9	30.7	33.8	36.0	35.6	36.6	35.2	32.4	28.5	26.3	(19)
(20)			MCWB		20.2	21.2	20.8	21.7	23.7	24.7	25.6	25.4	24.6	23.4	21.9	21.6	(20)
(21)		2%	DB		23.3	24.6	26.6	29.1	32.2	34.1	34.4	35.0	33.5	30.8	26.9	24.6	(21)
(22)			MCWB		19.4	20.2	20.3	21.8	23.4	24.8	25.7	25.6	24.4	23.5	21.3	20.8	(22)
(23)		5%	DB		22.0	23.2	25.3	28.0	31.1	33.0	33.5	34.0	32.5	29.5	25.2	22.9	(23)
(24)			MCWB		18.8	19.6	19.9	21.2	23.3	24.9	25.7	25.7	24.4	22.5	20.1	20.1	(24)
(25)		10%	DB		20.4	22.0	24.0	26.8	30.0	32.1	32.6	32.9	31.3	28.0	23.7	21.4	(25)
(26)			MCWB		17.8	18.8	19.4	20.6	23.0	24.7	25.6	25.6	24.1	21.9	19.6	18.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		21.8	22.6	22.8	24.5	26.3	27.5	27.5	27.8	26.9	26.1	24.0	22.9	(27)
(28)			MCDB		23.5	24.3	25.6	27.9	30.2	31.2	32.0	32.5	30.7	28.8	26.3	24.9	(28)
(29)		2%	WB		20.7	21.6	21.9	23.5	25.3	26.8	26.9	27.0	26.3	25.2	22.8	21.8	(29)
(30)			MCDB		22.5	23.6	24.3	26.6	29.1	30.7	31.5	31.7	30.0	28.3	24.9	23.5	(30)
(31)		5%	WB		19.6	20.7	21.3	22.8	24.6	26.3	26.6	26.7	25.7	24.5	21.8	20.8	(31)
(32)			MCDB		21.1	22.4	23.8	25.8	28.5	30.3	31.2	31.4	29.6	27.6	24.1	22.7	(32)
(33)		10%	WB		18.2	19.6	20.5	22.2	24.1	25.7	26.2	26.2	25.2	23.6	20.6	19.3	(33)
(34)			MCDB		20.0	21.3	23.1	25.2	27.9	29.7	30.6	30.9	29.1	26.5	23.2	21.0	(34)
(35)	Mean Daily Temperature Range		MDBR		10.0	9.8	10.3	10.5	9.8	9.0	8.5	9.1	9.7	11.1	11.2	10.2	(35)
(36)		5% DB	MCDBR		9.8	9.6	9.8	10.1	10.2	10.0	9.4	10.1	10.5	11.0	11.3	10.2	(36)
(37)			MCWBR		6.7	6.0	5.1	4.5	3.5	3.1	3.1	3.2	3.6	4.7	6.4	7.1	(37)
(38)		5% WB	MCDBR		8.7	8.3	8.4	8.4	8.8	8.6	8.6	9.1	8.7	9.0	9.8	9.3	(38)
(39)			MCWBR		7.3	6.4	5.7	4.8	3.5	3.5	3.3	3.4	3.5	4.8	7.1	7.5	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.337	0.354	0.378	0.420	0.447	0.475	0.494	0.487	0.446	0.380	0.357	0.350	(40)
(41)		taud			2.506	2.466	2.387	2.276	2.235	2.201	2.183	2.200	2.283	2.444	2.455	2.473	(41)
(42)		Ebn at Noon			896	910	906	876	849	821	804	807	828	870	863	859	(42)
(43)		Edh at Noon			89	101	117	135	142	146	148	144	128	102	93	89	(43)
(44)	All-Sky Solar Radiation	RadAvg			2.90	3.38	4.62	5.64	6.20	6.07	5.83	5.51	5.06	4.46	3.36	2.59	(44)
(45)		RadStd			0.28	0.50	0.41	0.41	0.37	0.51	0.43	0.49	0.45	0.52	0.32	0.27	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+105	(46)
(47)	Regional (26 neighbors)	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+119	+81	(47)

Nomenclature: See separate page